

Q3. (40 pts) A binary tree is a fascinating data structure with many variations, including binary search trees, AVL trees, red-black trees, complete binary trees, and max/min heaps. These variations can be classified as shape-based (structural constraints) or criteria-based (rules such as ordering). Choose one shape-based tree and one criteria-based tree, and provide a brief description of each.

A3: 1. shape-based: complete binary tree

a. 除最後一層外, tree 的每層都需填滿;
最後一層需盡可能向左排列

b. 可以使用簡單以 array 表示, 不會在記憶體豐中
留下空間

2. Criteria-based: binary search tree

a. 對每個 Node N

Left subtree 所有 node 的值小於 N

Right subtree 所有 node 的值大於 N

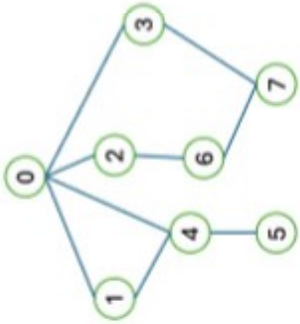
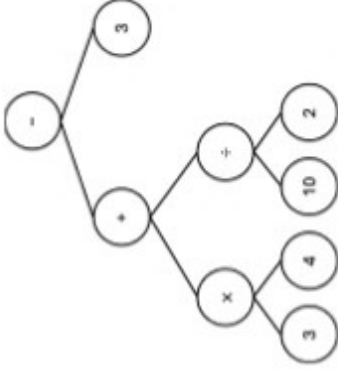
b. search, insert, delete 等操作 time complexity
為 $O(\log n)$, 效能好

Due: December 16, 2025, 17:00 (Room R1102)

Important Notice: You must print this take-home quiz and **write your answers by hand with a pen.**

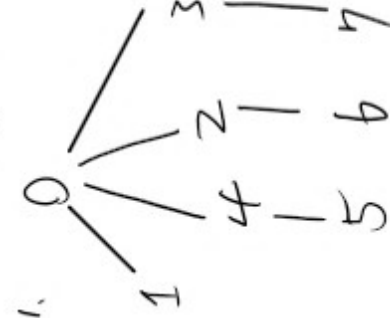
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Q1 Figure	Q2 Figure
	

Q1. (30 pts) Explain Breadth-First Search (BFS) on the graph and provide the BFS traversal order for the graph shown in Q1 Figure.

A1:



2. $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 6 \rightarrow 7$
 $\rightarrow 5$

Q2. (30 pts) In tree traversal, one common method is inorder traversal. Please use inorder traversal to print the arithmetic expression represented by the expression tree in Q2 Figure, and then evaluate it to compute the final result.

A2: 1. $3 \times 4 + 10 \div 2 - 3$

2. 14